DON BOSCO INSTITUTE OF TECHNOLOGY

Department of Computer Science and Engineering

under DBIT_SIC

AUTOMATED ACCIDENT DETECTION AND ALERT SYSTEM

SAMSUNG Innovation Campus Smart Things Edition

Problem statement

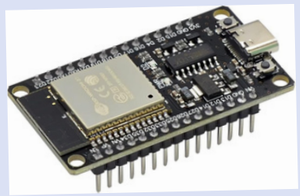
Develop an IoT- based automated accident detection and alert system that addresses delay in accident detection which leads to loss of lives due to lack of timely help. This system detects accident using gyroscope sensor and tracks the location using NEO-6M GPS module, sending the live location to gaurdians via SMS.

Objectives

Detect accidents in real-time and tracks GPS location of the accident instantly.

Logs accident data to ThingSpeak for analysis and Send automatic alerts via SMS using Twilio.

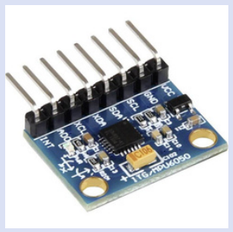
Components used



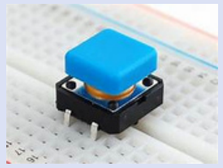
ESP32 Microcontroller



NEO 6M GPS module



MPU6050 Gyroscope



Push button



Buzzer

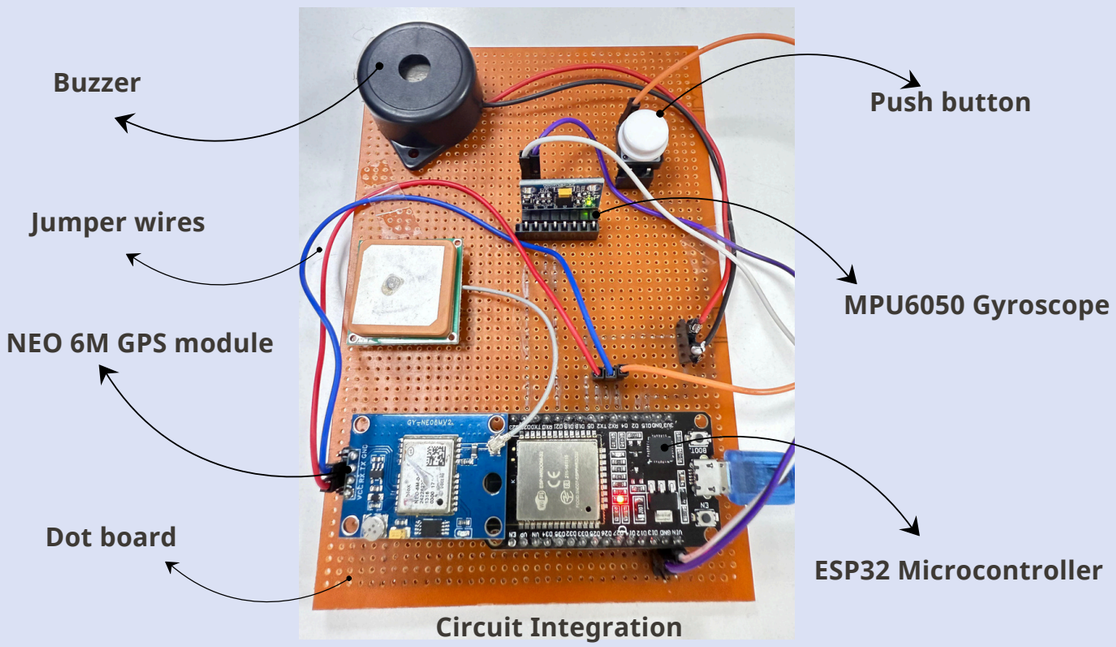
Software used

ARDUINO

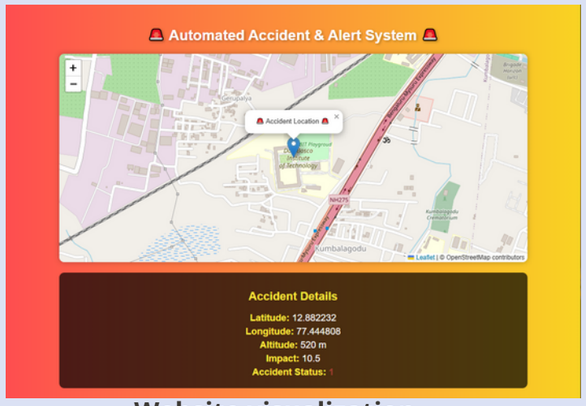
ThingSpeak™

twilio

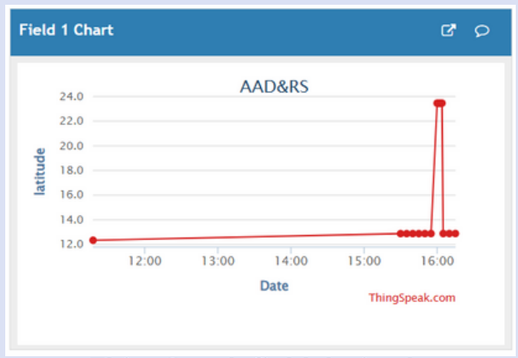
Working



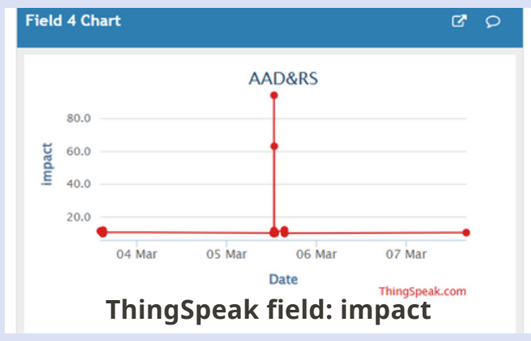
Circuit Integration



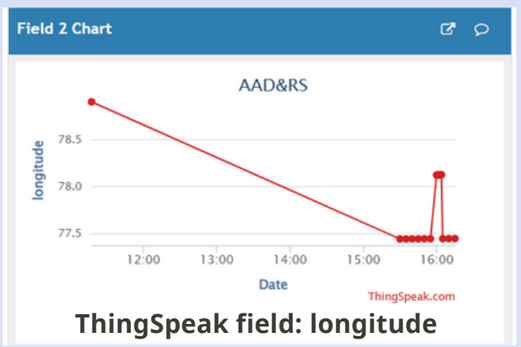
Website visualization



ThingSpeak field: latitude



ThingSpeak field: impact



ThingSpeak field: longitude

Operational workflow

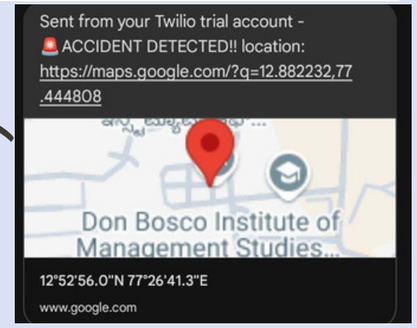
- ESP32 microcontroller processes the data from the gyroscope sensor and gps module.
- When the impact crosses threshold impact: triggers buzzer waiting for driver to abort mission.
- If the driver doesn't press button within 10 seconds, the alert SMS is sent to the guardian along with gps location.
- Simultaneously the data is sent to the ThingSpeak cloud and updated in the website for analysis and visualisation.

Test cases and output

Case 1: button is not pressed

```
Impact: 10.58
Accident status: 1
⚠️ Impact Detected! Waiting for driver response...
🚨 No response! Activating buzzer...
📡 Sending accident alert...
accidentStatus and impact sent successfully!
📩 SMS Sent Successfully!
```

Twilio alert SMS



Case 2: button is pressed

```
Impact: 10.20
Accident status: 1
⚠️ Impact Detected! Waiting for driver response...
🚗 False Alarm! Alert Aborted.
```

Conclusion

By integrating sensors, GPS tracking, and instant messaging, this IoT project ensures quick emergency response, potentially saving lives. The system is cost-effective, efficient, and can be further improved for real-world deployment.

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